PDS data selection

Trigger activities and scintillation signatures

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PDS TAs: live readout and replay builders

Capability	Evidence
Live prescaled TAs	Run 45392: 29 matching 1 ms PDS + emulated-charge windows.
Coincidence builder	Replay: six distinct channels in 10 μs; 93,126 TAs.
Prompt-light builder	Replay-tested descriptor prompt/total integral proxy.

Live prescale fix: PDS TA type + one-tick sample units, via session-local interposer.

Coincidence and prompt-light algorithms tested in replay; live DAQ integration pending.

Scintillation motivates coincidence and light timing



Coincident channels | Light sum | Prompt / total

Thresholded descriptor sums are proxies.

Calibrated light and exact gate integration need waveforms.

Conceptual envelope, not a measurement. Scintillation reference.

Next: integrate and qualify physics TAs

INPUT

Merge TPs across links by timestamp

Common watermarks, quiet-source progress, duplicate/loss accounting.

SELECTION

Register a PDS TA plugin + algorithm ID

Configure region, channel multiplicity, gates and overlap policy.

VALIDATION

Compare against calibrated waveform observables

Measure efficiency, accidental rate, latency and sustained throughput.